

SERVICE MANUAL

Thermal Receipt Printer

RP-600



Partner Tech Corporation

Revision history

Data	Version	Description	Author
2013-4-2	V1.0	First release	Yu Yan, Sun Chuanliang, Jiang Jinhao

CONFIDENTIAL

Declaration

Information in this document can not be changed without notice. Partner Tech Corporation reserves the right to improve products as new technology, components, software, and firmware become available. If users need further data about this product, please feel free to contact Partner Tech Corporation or your local dealer.

No part of this document may be reproduced or transmitted in any form by any means, electronic or mechanical, for any purpose without the express written permission of Partner Tech Corporation.

Copyright

Copyright © 2013 by Partner Tech Corporation

Printed in China

Version 1.0

Trademark

Registered trademark of Partner Tech Corporation **PARTNER**

Warning and Caution

 **Warning:** Items shall be strictly followed to avoid injury or damage to body and devices.

 **Caution:** Items with important information and prompts for operating the printer.

Maintenance caution

- (1) Follow the steps in this manual during maintenance;
- (2) Make sure that the printer and the computer are turned off before connecting or disconnecting the communication cable, changing print head or doing maintenance to the printer;
- (3) Be sure to protect it against electrostatic damage when maintaining print head and other electronic components;
- (4) Time between turning on and turning off the printer should be no less than 20 seconds;
- (5) Do not print when no paper roll is installed. Doing so can damage platen roller and print head seriously;
- (6) Set the print darkness to a lower grade as long as the print quality is acceptable. This will help to keep the print head durable.
- (7) The content regarding the cutter only applies to models configured with cutter.

Content

1 PRINTER FEATURES	1
1.1 INTRODUCTION.	1
1.2 MAIN FEATURES.	1
1.3 TECHNICAL SPECIFICATIONS	1
1.4 SWITCH, BUTTON AND LED.	3
1.4.1 Power switch	3
1.4.2 Button	3
1.4.3 LED and buzzer	3
2 WHOLE PRINTER AND COMPONENT MODULES	5
2.1 PRINTER OVERVIEW.	5
2.2 CONNECTION DIAGRAM OF CONTROL BOARD MODULE	5
3 COMMUNICATION INTERFACE	7
3.1 RS-232 SERIAL INTERFACE.	7
3.1.1 Interface connector and specifications	7
3.1.2 Interface signal definition	7
3.1.3 Table 3.1-1 25 pin interface signal definition	7
3.1.4 Communication cable	7
3.1.5 Interface parameters	7
3.1.6 Default configuration of printer interface	8
3.1.7 Printer interface time sequence	8
3.1.8 Table 3.1-4 DTR/DSR signal	8
3.1.9 Troubleshooting of communication interface	8
3.2 PARALLEL INTERFACE	9
3.2.1 Interface connector and specifications	9
3.2.2 Interface signal definition	9
3.2.3 Communication cable	10
3.2.4 Interface parameters	10
3.2.5 Table 3.2-2 parallel interface parameters	10
3.2.6 Time sequence of printer interface	10
3.2.7 Troubleshooting of communication interface	10
3.3 JK-E05 ETHERNET INTERFACE.	11
3.3.1 Interface connector and specifications	11

3.3.2 Troubleshooting of communication interface	11
3.4 JK-W01 WIRELESS ETHERNET INTERFACE	12
3.4.1 Function and technical specifications	12
3.4.2 Function description	12
3.4.3 Electrical features	12
3.4.4 Network card parameter features.....	12
3.4.5 Troubleshooting of communication interface.....	13
4 DISASSEMBLY AND ASSEMBLY	14
4.1 REPAIR TOOLS	14
4.2 PRINTER DISASSEMBLY	14
4.2.1 Printer housing disassembly	14
4.2.2 Interface board disassembly	17
4.2.3 Disassembly of main control board and its protective film.....	17
4.2.4 Power switch disassembly	18
4.2.5 Guide plate disassembly.....	18
4.2.6 Interface baffle disassembly.....	18
4.2.7 Top cover holder disassembly.....	19
4.2.8 Disassembly of cutter sliding blade and print platen roller	20
4.2.9 Disassembly of cutter fixed blade and print head	21
4.2.10 Paper cabinet disassembly	22
4.2.11 Disassembly of paper sensor / mark sensor.....	24
4.2.12 Buzzer disassembly	24
4.2.13 Paper near end sensor disassembly	24
4.2.14 Motor disassembly.....	25
4.2.15 Gear disassembly.....	25
4.2.16 Micro switch disassembly.....	25
4.3 ASSEMBLY	25
5 ROUTINE MAINTENANCE.....	26
5.1 CLEANING PRINT HEAD AND PLATEN ROLLER	26
5.2 CLEANING PAPER END/MARK SENSOR.	26
6 TROUBLESHOOTING.....	27
6.1 TROUBLESHOOTING.	27
6.2 PRINTER ACTIONS WHEN AN ERROR OCCURS	27

6.3 PRINTER STATUS DESCRIPTION.	27
APPENDIX.....	29
APPENDIX 1 COMMAND LIST.	29
APPENDIX 2 EEPROM CONFIGURATION TABLE	29
APPENDIX 3 LIST OF EASILY DAMAGED PARTS.	30
APPENDIX 4 PRINTER EXPLODED VIEW	34
APPENDIX 5 PARTS LIST.	38
APPENDIX 6 OVERALL SIZE (UNIT: MM)	40
APPENDIX 7 SILK-SCREEN DRAWING OF MAIN CONTROL BOARD	41
APPENDIX 8 BUTTON CONFIGURATION LIST.	42

CONFIDENTIAL

1 Printer features

1.1 Introduction

RP-600 is developed for high-end thermal receipt printing market, which is widely used.

RP-600 can be connected with other devices via parallel, serial, USB, Ethernet, or WLAN interface. It provides drivers for operating systems such as Windows 2000/XP/Server2003/VISTA/Server2008/WIN7/WIN8, POSReady2009/POSReady7, Linux, Mac, and UPOS middleware.

Note: Windows and Linux drivers support 64 bit operating system.

1.2 Main features

- Low noise, high printing speed
- Easy paper loading
- Easy operation and maintenance
- Support marked paper and continuous paper
- Compatible with various paper widths between 57.5~82.5mm
- Cutting paper automatically
- Cash drawer control interface
- Communication interface optional
- Compatible with ESC/POS commands
- Volume of buzzer adjustable
- Paper saving function
- Low power consumption design

1.3 Technical specifications

Item		Parameter
Printing	Print method	Direct thermal line printing
	Resolution	180 × 180 DPI
	Paper width	57.5~82.5mm, continuously adjustable
	Print width	Max. 72mm
	Print speed	Standard: Max.230mm/s Gray scale: Max.150mm/s
	Flash LOGO	Max. 512K bytes
	Data buffer	64K bytes, 4K bytes or 45 bytes
Interface	Fixed USB on board	USB2.0 (full speed)
	Optional interface	Serial (RS-232C) Parallel (IEEE1284) Ethernet WLAN
	Cash drawer	Controlling 1~2 cash drawers can be selected
Memory		RAM: 2MB, Flash: 2MB/4MB
Printer status detection		Paper end sensor/ black mark sensor, paper near end sensor, top cover

		position sensor, print head temperature sensor, printer voltage detection
Cuter mode		Full cut, partial cut, tear-off
Barcode	1D	UPC-A, UPC-E, JAN8 (EAN8), JAN13 (EAN13), CODE 39, CODE 93, CODE 128, ITF, CODABAR, GS1-128, GS1 DataBar
	2D	PDF417, QR, Maxicode, 2D GS1 DataBar, Composite Symbology
Fonts		Font A: 12 × 24 Font B: 9 × 17 Kanji font A: 24 × 24
Character set		95 Alphanumeric
		14 types of international characters
		128 x 68 code page
		Traditional Chinese, Simplified Chinese (GB2312/18030), Japanese, Korean, English, HK (optional)
		User-defined font (95) & code page
Character enlargement		All characters can be enlarged 1-6 times horizontally and vertically
Character rotation		Rotation printing in four directions (0°, 90°, 180°, 270°)
Command		ESC/POS compatible
Paper	Paper type	Thermal continuous paper, Thermal marked paper
	Paper roll OD	Max. 83mm
	Paper thickness	0.06mm~0.10mm
Power supply	Input	100-240VAC, 50-60Hz
	Output	24V ± 5% DC, average current 2.15A
	External only	Power adapter
Human-machine interface		Power switch Micro switch
		Button FEED button
		LED POWER LED, ERROR LED
		Buzzer Integrated internal buzzer, the volume can be adjusted
Reliability	Lifetime of print head	≥150Km (standard test sample with 12.5% duty ratio)
	Lifetime of cutter	1,500,000 cuts (under 0.063mm paper thickness)
	MCBF	60,000,000 lines
	MTBF	360,000 hours (main control board)
Operation temperature and humidity		5°C to 45°C, 20% to 90%RH (40°C)
Storage temperature and humidity		-40°C to 60°C, 20% to 93%RH (40°C)
Overall dimensions		196mm×146mm×148mm (L×W×H)
Functions	Saving paper	Support
	Feed button configuration	Support (configure the printer without computer)
	Right-up-side printing	Support
	Water mark printing	Support
	Gray scale printing	Support

Table 1.3-1 technical specifications

1.4 Switch, button and LED

1.4.1 Power switch

On the right of the printer, press “O” to turn off the power and press “—” to turn on the power.



Figure 1.4-1 power switch

1.4.2 Button

1) Under power-on self-test status:

Turning on the power switch while pressing the FEED button, the printer will print configuration sample, cut paper, and enter hold-on status (error LED flashing). If pressing the FEED button momentarily at this time, the printer will print out character test sample and enter normal standby status after the printing.

2) Enter button configuration mode:

Turning on the power switch while pressing the FEED button, the printer will print configuration sample, cut paper, and enter hold-on status (error LED flashing). If pressing the FEED button for a long time at this time, the printer will enter button configuration mode (set the printer parameters offline through FEED button). For the detailed functions and operating methods of button configuration mode, refer to appendix 8 Button Configuration List.

3) Under standby status:

The printer feeds paper continuously if pressing down the FEED button, and stops feeding paper after releasing the FEED button.

4) Under cutter error status:

If pressing the FEED button momentarily, the printer will try to clear the cutter error automatically.

5) During the printing:

The printer has no operation after pressing down the FEED button.

1.4.3 LED and buzzer

➤ Function description of LED and buzzer

LED name	Status	Description
Power LED green	Always on	The printer is on.
	Off	The printer is off.
Error LED red	Off	The printer is in normal status.
	Flash	The printer is in error status or paper near end status.
Buzzer	Beep	The printer is in error status.

Table 1.4-1 function description of LED and buzzer

➤ Description of LED and indication information

Error type	Error LED	Buzzer
Print head overheating	Cycle flash 6 times	Cycle beep 6 times
Printer voltage abnormal	Cycle flash 5 times	Cycle beep 5 times
Cutter error	Cycle flash 4 times	Cycle beep 4 times
Print head lift-up	Cycle flash 3 times	Cycle beep 3 times
Paper end	Cycle flash 2 times	Cycle beep 2 times
Paper near end	Quick cycle flash	No beep
Mark missing or calibration failure	Slow cycle flash	No beep

Table 1.4-2 description of LED and indication information

 Caution:

The printer detects the temperature of print head using the thermal resistor. If print head is overheating, the protective circuit will cut off the power of print head and stop the printing. The threshold temperature when the print head stops the printing is 65℃。

2 Whole printer and component modules

2.1 Printer overview

RP-600 consists of printer and power adapter.

Printer mainly includes three modules: 1 housing module, 2 mechanism module, and 3 main control board module.

Modules are disassembled as follows:

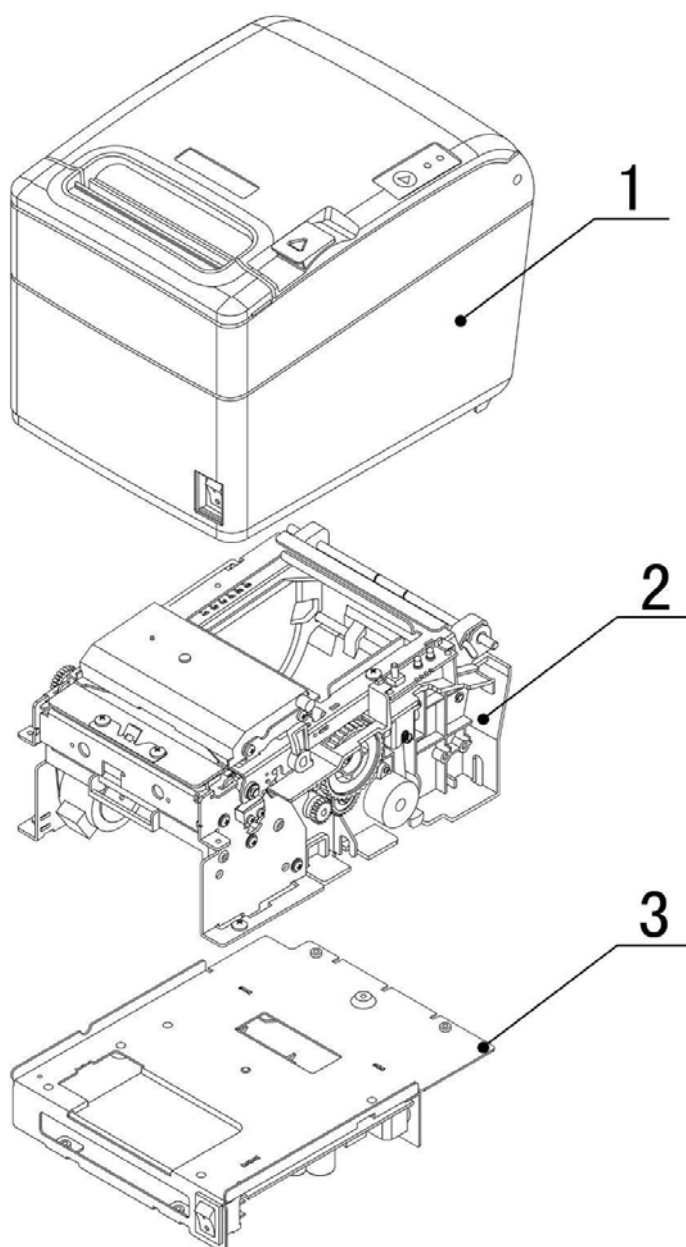


Figure 2.1-1 printer components

2.2 Connection diagram of control board module

The printer uses POSMV1.3 main control board, and sensor, button, motor, and cutter, etc are connected to the main control board via connector or pinboard. The positions of sockets on the main control board are as follows:

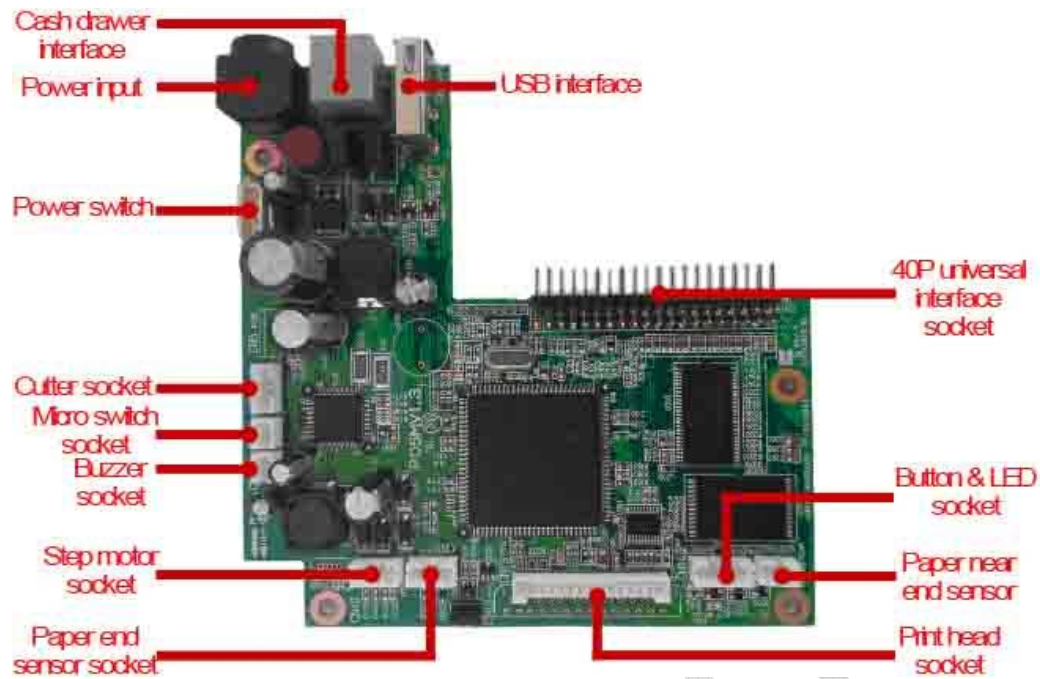


Figure 2.2-2 positions of sockets on the main control board

3 Communication interface

3.1 RS-232 serial interface

3.1.1 Interface connector and specifications

The serial interface of the printer is compatible with RS-232C standard, the socket of which is DB25F.

3.1.2 Interface signal definition

25 pin interface signal is defined as below:

Pin	Signal name	Signal direction	Function
1	FG	—	Frame ground
2	TXD	Output	Transmit data
3	RXD	Input	Receive data
5	NC	—	Not connected
6	DSR	Input	Data set ready
7	SG		Signal ground
8—19	NC	—	Not connected
4, 20	DTR	Output	Data terminal ready
21—25	NC	—	Not connected

Table 3.1-1 25 pin interface signal definition

3.1.3 Communication cable

The communication cable of 25 pin serial interface is 9 to 25 serial interface cable.

Host	Printer
2 RXD-----	2 TXD
3 TXD-----	3 RXD
8 CTS -----	4 RTS
7 RTS -----	5 NC
4 DTR-----	6+8 DSR
1+6 DSR-----	20 DTR
5 SG -----	7 SG

3.1.4 Interface parameters

The interface parameters are as follows:

Item	Parameter
Data transmission	Serial
Synchronous mode	Asynchronous transmission
Handshake mode	DTR/DSR or XON/XOFF control
Level	MARK = -3 to -15 V: logic 1/ OFF SPACE = +3 to +15 V: logic 0/ ON
Baud rate	1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200 bps
Data bit	8 bits
Stop bit	1 bit

Table 3.1-2 interface parameters

Note: The handshake mode, baud rate and other parameters of serial interface can be set by EEPROM.

3.1.5 Default configuration of printer interface

The default configuration of printer interface is as below: (users can query the related configuration of interface by printing the configuration sample)

Baud rate: 115200bps

Data bit: 8 bits

Parity: None

Stop bit: 1 bit

Handshake mode: DTR / DSR

3.1.6 Printer interface time sequence

- Printer software handshake (XON / XOFF) time sequence:

When XON/XOFF control mode is selected, the printer transmits XON/XOFF signal as below:

	Printer status	Action
XON transmission	1) When the printer is turned on and enters connection status.	Transmit
	2) When the receive buffer is released from full status.	Transmit
XOFF transmission	1) The receive buffer becomes full.	Transmit

Table 3.1-3 XON/XOFF signal

Note:

1. Definition of buffer full: When the interface is serial interface and the space of buffer is lower than a certain number of bytes (number of bytes is n1), the command buffer is full and keeps full. It cannot get out of the full status until the buffer space increases to a certain number of bytes (number of bytes higher than n1).
2. Definition of XON / XOFF: XON character is 0x11; XOFF character is 0x13.

- Printer hardware handshake (DTR/DSR) time sequence:

When the hardware handshake is selected, the DTR / DSR signal status of printer is as follows:

	Printer status	Status
DTR	1) When the printer is turned on and enters connection status.	SPACE
	2) When the receive buffer is released from full status.	SPACE
DSR	1) The receive buffer becomes full.	SPACE

Table 3.1-4 DTR/DSR signal

3.1.7 Troubleshooting of communication interface

Malfunction: Data cannot be sent or received.

Causes:

1. Connecting wires are not connected reliably;
2. Parameters of serial interface are not set correctly;
3. Some components of serial interface circuit are welded incompletely or damaged.

Troubleshooting:

- 1) Check whether the serial interface cable is connected reliably. If not, reconnect it.
- 2) Check whether serial interface settings of host are consistent with the configuration sample of printer. If not, reconfigure the serial interface settings of host, and resend the data for test.
- 3) Check whether the serial communication signal is normal or not. If not normal, check whether the signal channel is normal or not; if normal, check whether the serial interface board RS232 conversion chip is welded incompletely or not; if not welded incompletely, replace this chip.

3.2 Parallel interface

3.2.1 Interface connector and specifications

The parallel interface connector is Centronics 36 pin parallel interface connector.

3.2.2 Interface signal definition

36 pin interface signal is defined as below:

Pin No.	Signal source	Signal definition
1	H	Data latching pulse, data is latched into the printer on the rising edge of the pulse.
2	H	Data 0(least significant bit)
3	H	Data 1
4	H	Data 2
5	H	Data 3
6	H	Data 4
7	H	Data 5
8	H	Data 6
9	H	Data 7 (most significant bit)
10	P	Printer ACK signal, indicating that the printer has received the data of the previous byte, which is a negative pulse lasting for about 1us.
11	P	Printer is busy. High level means printer can not receive data any more.
12	P	Paper end signal, high level means paper end.
13	P	Voltage is pulled up to 5V by a resistor.
14	—	Not connected
15	—	Not connected
16	Logic ground	
17	Frame ground, isolated with logic ground.	
18	—	Voltage is pulled up to 5V by a resistor.
19~30	Logic ground	
31	H	Voltage is pulled up to 5V by a resistor.
32	P	Printer error signal, low level means that an error occurs. Printer outputs error signal together with paper end signal.
33	Logic ground	
34~35	—	Not connected
36	Logic ground	

Table 3.2-1 36 pin interface signal definition

Note: H means the signal is from host; P means the signal is from printer.

3.2.3 Communication cable

Communication cable is 25 to 36 parallel interface cable.

3.2.4 Interface parameters

Parallel parameters are as follows:

Item	Parameter
Data transmission	8 bit parallel
Synchronous mode	Exterior nStrobe signal
Handshake mode	BUSY signal
Signal level	TTL compatible
Status transmission	PE (paper status), FAULT (printer status)

Table 3.2-2 parallel interface parameters

3.2.5 Time sequence of printer interface

Sequence diagram of parallel interface receiving data:

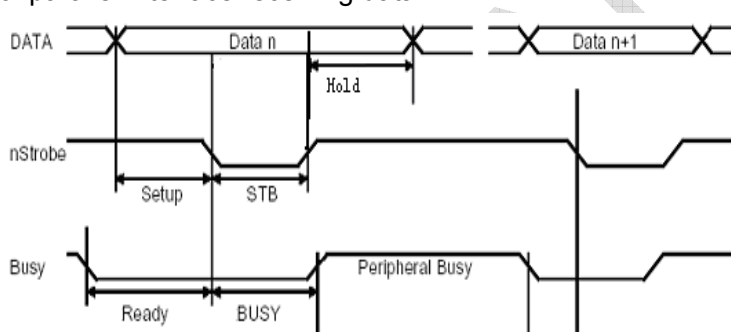


Figure 3.2-1 sequence diagram of parallel interface receiving data

Signal	Min.	Max.
setup	0.75 us	-
ready	-	-
stb	0.75 us	500 s
busy	-	0.5 us
hold	0.75 us	-

Table 3.2-3 time sequence of parallel interface receiving data

3.2.6 Troubleshooting of communication interface

Corresponding relationship between printer status and signal wire:

Status	FAULT	PE
Normal	High	Low
Paper end	Low	High
Print head overheating	Low	Low
Top cover open	Low	Low

Table 3.2-4 corresponding relationship between printer status and signal wire

When the errors above occur, users can read out the printer status via parallel interface.

Malfunction: Data can not be sent or received.

Causes:

- 1) Cables are not connected securely;
- 2) Parallel interface circuit is incompletely welded or damaged.

Troubleshooting:

- 1) Check if parallel cables are securely connected or not. If not, connect them securely.
- 2) Check if the parallel communication signal is normal or not. If not normal, check signal channel. If the signal channel is normal, check if any part of the parallel interface board is incompletely welded. If not, replace the main chip.

3.3 JK-E05 Ethernet interface

3.3.1 Interface connector and specifications

Meet the 10/100M BASE-T standard of IEEE802.3.

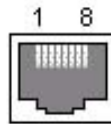


Figure 3.3-1 interface connector

Pin	Signal name	Description
1	TX+	Data transmission +
2	TX-	Data transmission -
3	RX+	Data receiving +
4	NC	Reserve
5	NC	Reserve
6	RX-	Data receiving -
7	NC	Reserve
8	NC	Reserve

Table 3.3-1 interface module pin list

3.3.2 Troubleshooting of communication interface

Malfunction: Unable to connect to the network.

Causes:

- 1) Connecting wires are not connected securely.
- 2) Some components of interface board are welded incompletely or damaged.

Troubleshooting:

- 1) Check whether the network cable is connected reliably or not. If not, reconnect it.
- 2) Check whether the interface board network LED is normal or not. If not normal, check the signal channel; if the signal channel is normal, check whether the related components on interface board are welded incompletely or not; if no incomplete welding exists, replace the main chip.

3.4 JK-W01 wireless Ethernet interface

3.4.1 Function and technical specifications

The interface module supports 802.11b, 802.11g and 802.11n protocols. The printer supports two print methods: 9100 port and LPR/LPD, and supports printer status back and the configuration and maintenance of interface module.

Item		Parameter
Communication interface		In accordance with 802.11b, 802.11g and 802.11n wireless standards
Supported basic communication protocol		ARP, RARP, IP, ICMP, TCP, UDP
Network card type		USB card
Communication speed		Max. communication speed is up to 150M Bps.
Basic function	Print function	Use 9100 port, LPR/LPD print
	Configure IP	Use DHCP, HTTP protocols
	Status back	Use 4000 port (TCP)
	Firmware update	Use TFTP protocol
Power	Input	Use the 5V power provided by main control part, current: no larger than 500mA
Environmental requirements	Operating environment	+5~45℃, 20-90%RH (no condensation)
	Storage environment	-40~60℃, 20-93%RH (no condensation)
Physical feature	Overall size	117(W)×79(D)mm
	Weight	About 93g

Table 3.4-1 technical specifications

3.4.2 Function description

USB interface is directly controlled by CPU and is connected with external LAN via wireless network card.

3.4.3 Electrical features

The interface uses wireless network card, and supports 802.11b/g/n protocol.

3.4.4 Network card parameter features

- Interface type: USB2.0
- Complied standards: compatible with Wi-Fi, IEEE802.11b/g/n
- Frequency channel: 1~13 channels
- Transmission speed varies with operating environment.
- Transmission distance: indoors without obstruct: 100m, outdoors without obstruct: 300m, varying with operating environment.
- Security: WSSID, WEP (64/128 bit encrypted)
WPA-PSK, WPA EAP (802.1x certification)
- Power supply: DC3.3V, max. current consumption 450mA
- System requirements: Windows XP/Vista/7/MAC.OS/Linux

3.4.5 Troubleshooting of communication interface

Malfunction: Unable to connect to the wireless network or unable to find the wireless network.

Causes:

- 1) Wireless network card is not plugged in or plugged in insecurely.
- 2) Some components of interface board are welded incompletely or damaged.

Troubleshooting:

- 1) Check whether the network card is plugged reliably or not. If not, reconnect it.
- 2) Check whether the wireless interface board LED is normal or not. If not normal, check the signal channel; if the signal channel is normal, check whether the related components on the wireless interface board are welded incompletely or not; if no incomplete welding exists, replace the main chip.

CONFIDENTIAL

4 Disassembly and assembly

Caution during operation:

- 1) Do not disassemble any parts of the printer or loosen any screw if it works properly;
- 2) When disassembling parts, please check carefully if the connecting wires are damaged or not;
- 3) When handling the print head or electronic components, please make sure to take antistatic measures;
- 4) During the disassembly, be careful not to leave parts like screw inside the printer;
- 5) Don't scratch the print head surface during assembly.

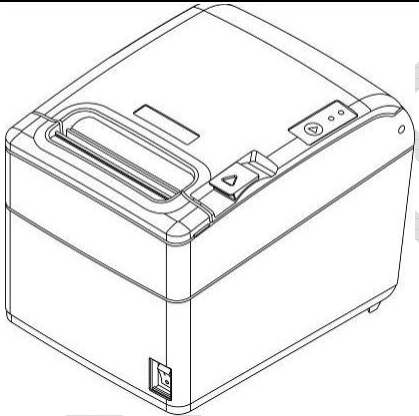
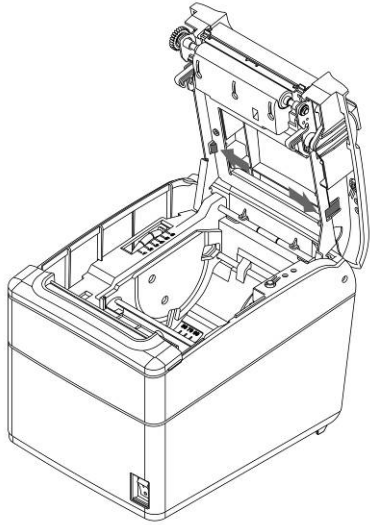
Assistant materials: grease, alcohol, absorbent ball.

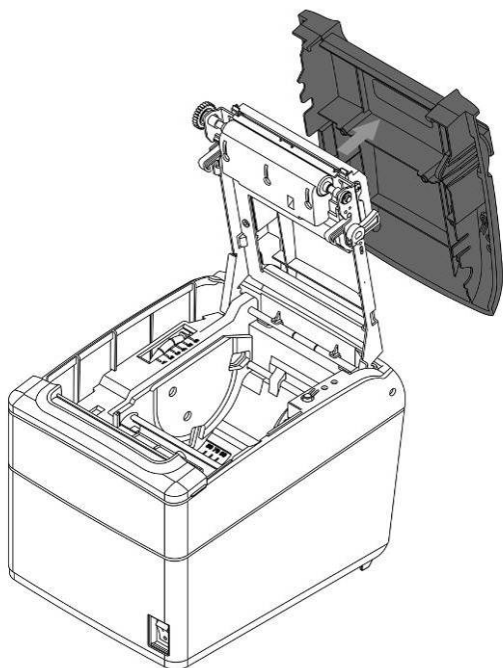
4.1 Repair tools

- Cross screw driver
- Nipper pliers
- Cutting pliers

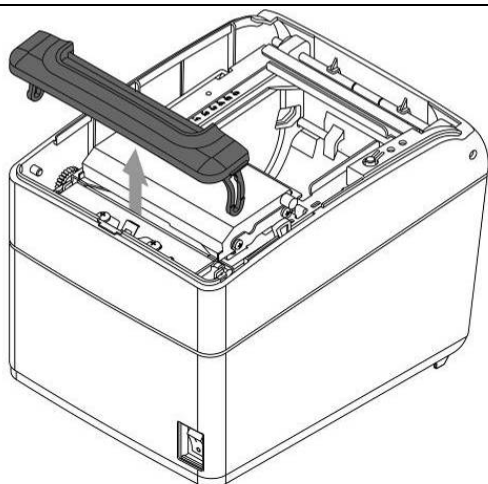
4.2 Printer disassembly

4.2.1 Printer housing disassembly

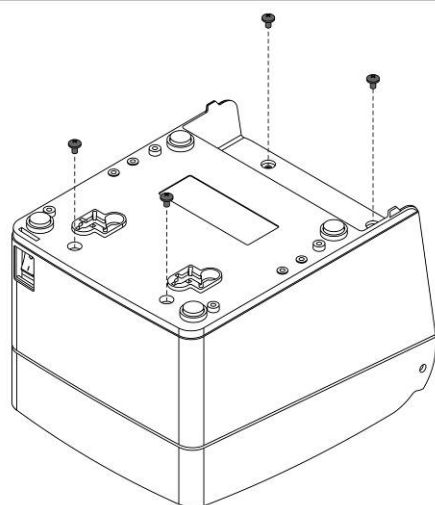
Figure	Description
	Printer isometric view
	Push the hooks on the left and right of the top cover in the direction shown in the figure to separate the hooks from the top cover holder.



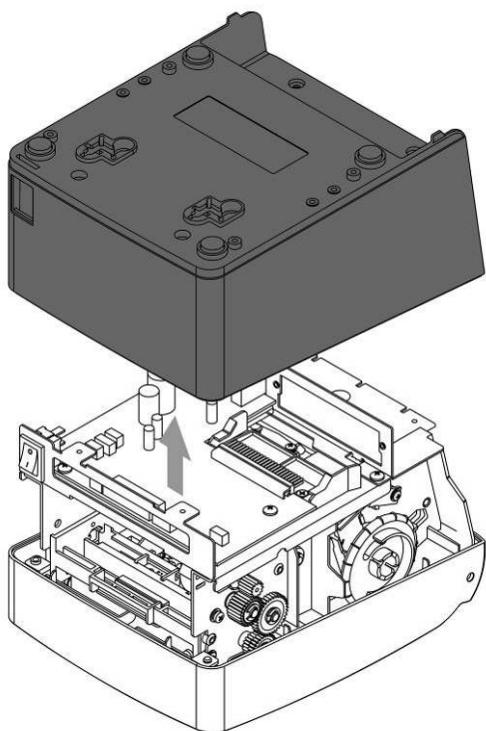
Take off the top cover.



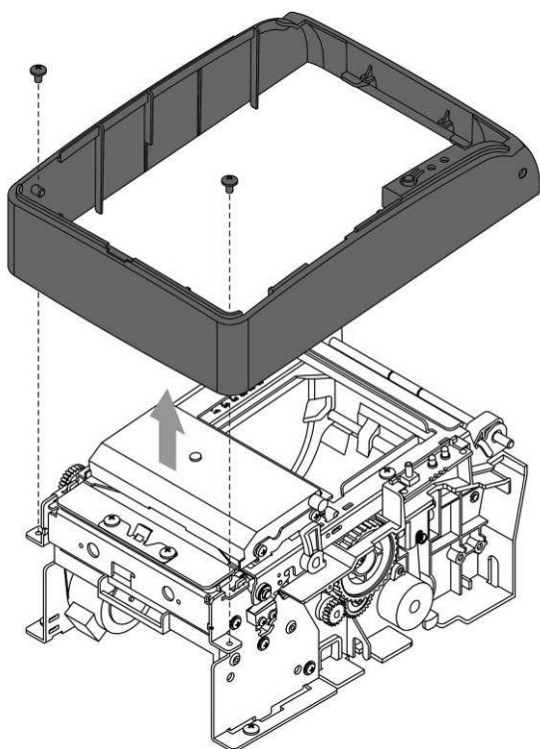
Lift up the printer cutter cover along both sides in the direction of the arrow, and take off the cutter cover.



Disassemble the four M3*5 screws fixing the bottom cover.

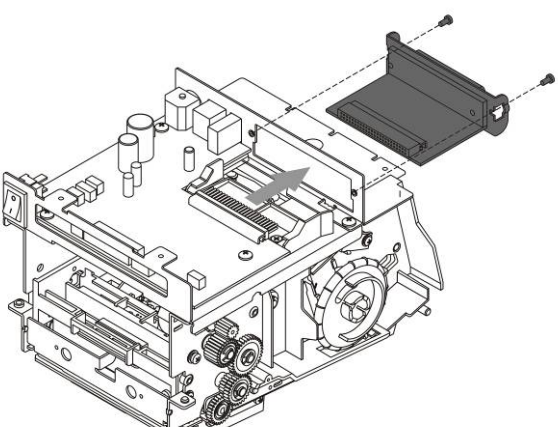


Take off the bottom cover.

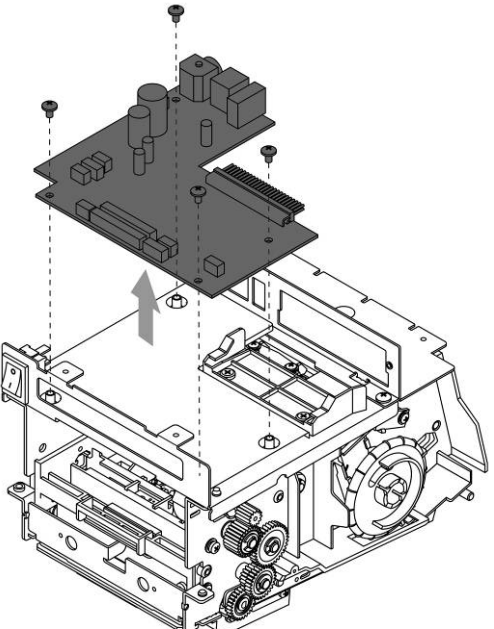
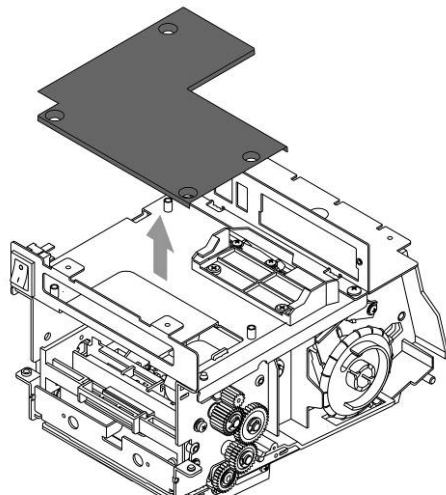


Disassemble the two M3*5 screws fixing the middle cover, and take off the middle cover in the direction of the arrow to complete the disassembly of printer housing.

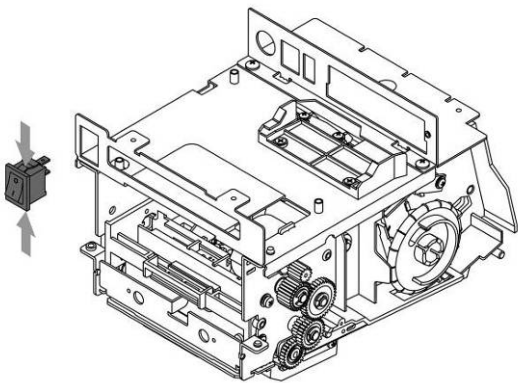
4.2.2 Interface board disassembly

Figure	Description
	Disassemble the two M2.5*4 pan head screws fixing the interface baffle and pull the interface board out from the back of the printer in the direction of the arrow.

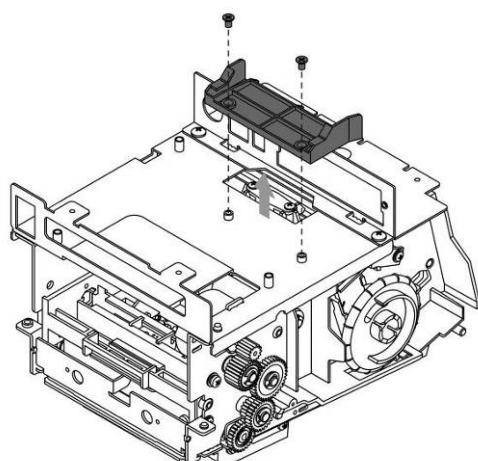
4.2.3 Disassembly of main control board and its protective film

Figure	Description
	Unplug the cables on sockets of main control board, disassemble the four M3*5 screws fixing main control board, and separate the main control board from the mechanism.
	Take off the main control board protective film in the direction of the arrow.

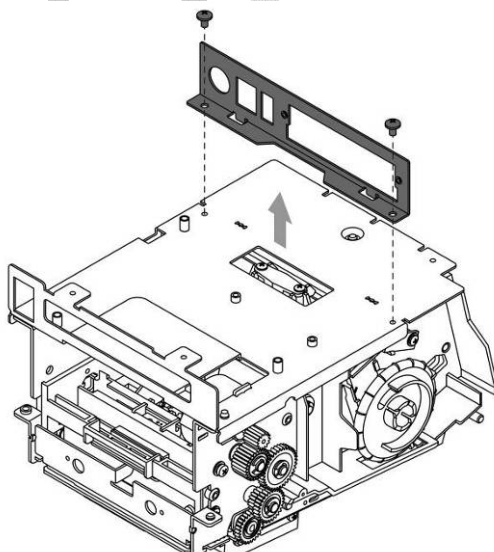
4.2.4 Power switch disassembly

Figure	Description
	Press and hold the flat spring of power switch in the direction of the arrow and take off the power switch from the cover plate of main board.

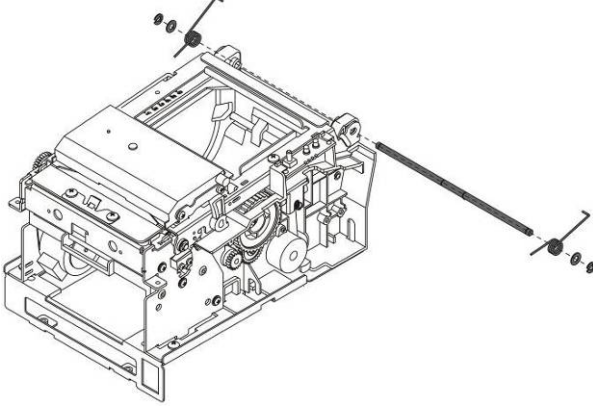
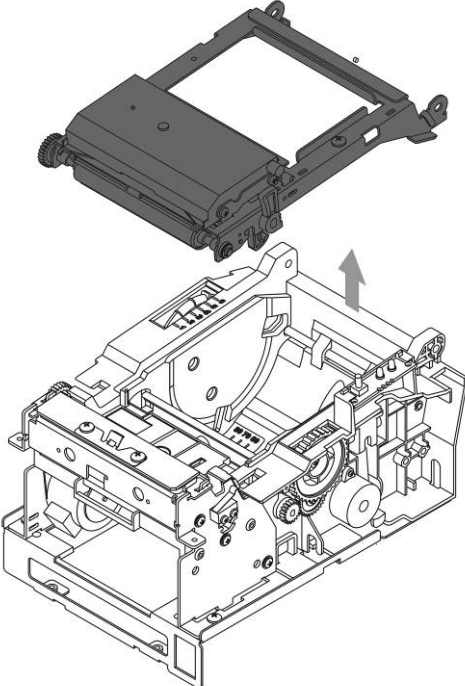
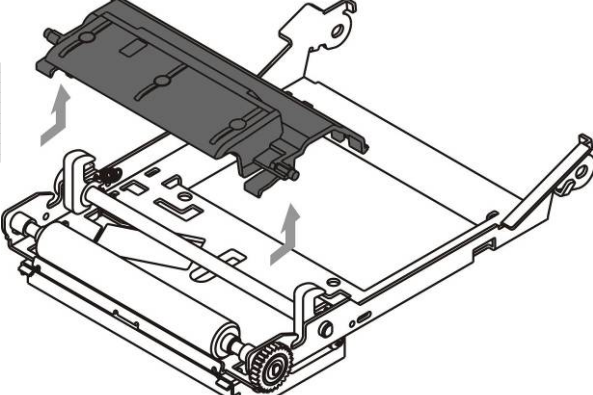
4.2.5 Guide plate disassembly

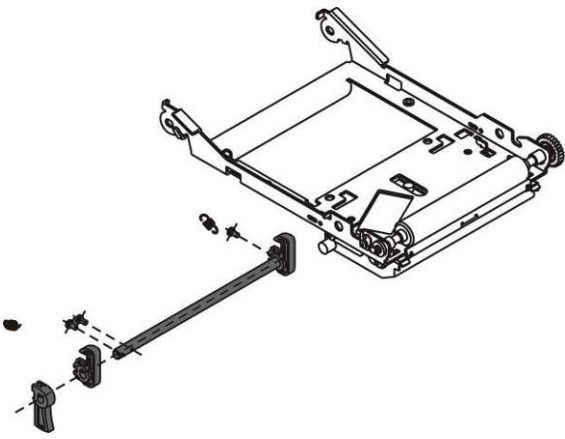
Figure	Description
	Disassemble the two M3*4 flat screws fixing the guide plate and take off the guide plate from the cover plate of main board.

4.2.6 Interface baffle disassembly

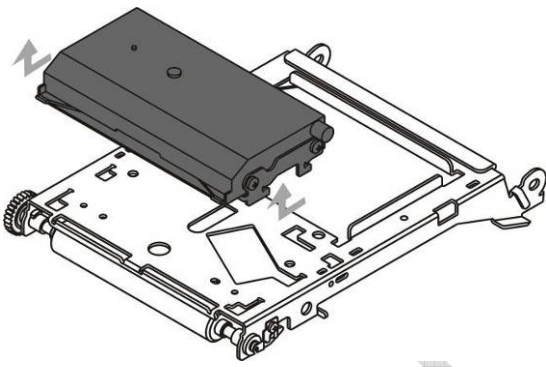
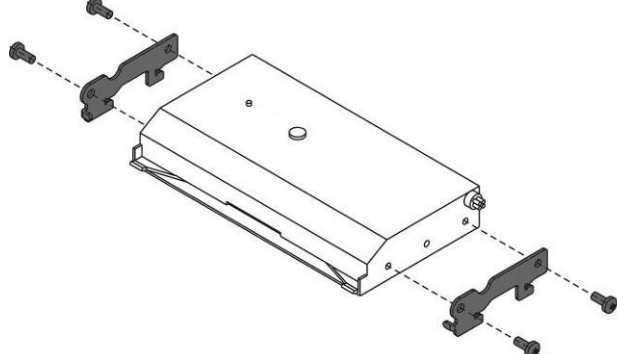
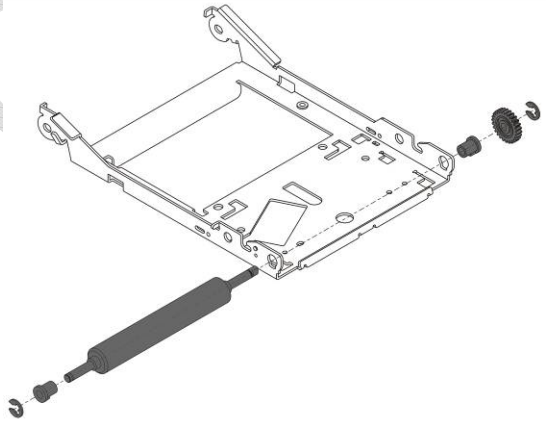
Figure	Description
	Disassemble the two M3*5 screws fixing the interface baffle and take off the interface baffle in the direction of the arrow.

4.2.7 Top cover holder disassembly

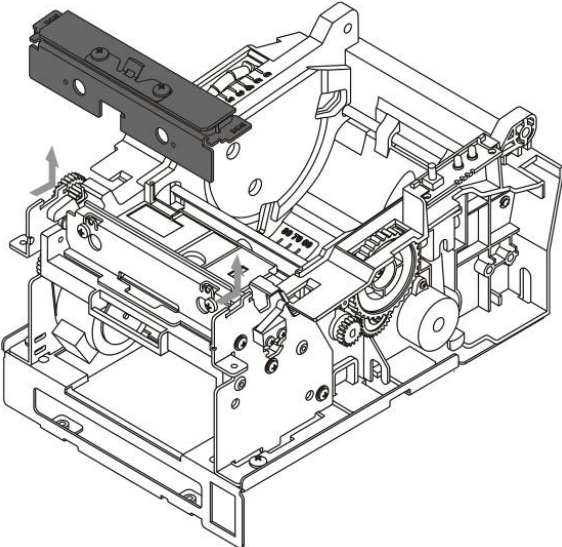
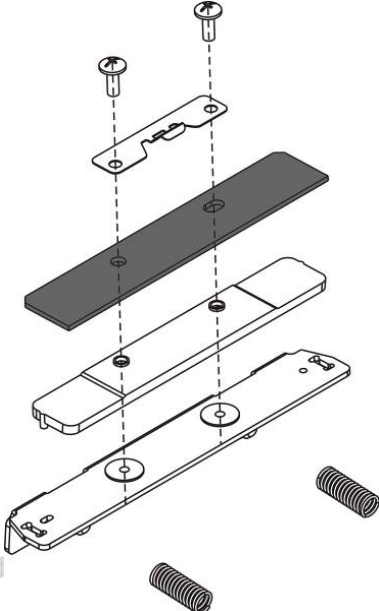
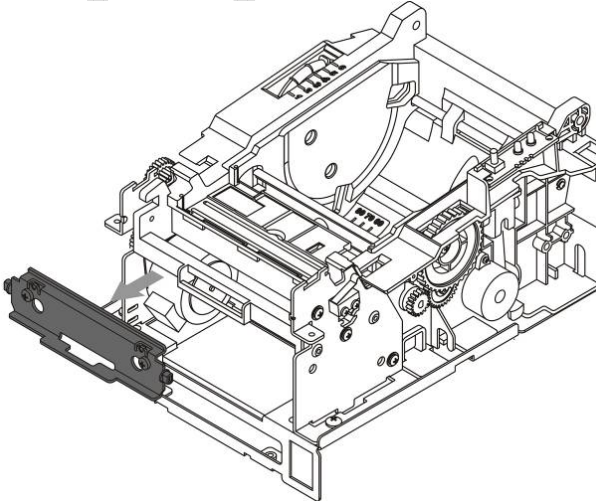
Figure	Description
	Disassemble the two $\varnothing 3$ "E"-rings fixing the top cover shaft, take off the $\varnothing 4$ flat gaskets and shaft torque springs on both sides in order, and pull the top cover shaft out.
	Take off the top cover holder from the mechanism, and complete the disassembly of top cover holder.
	Take off the upper mark sensor holder with force in the direction shown in the figure without using any tools.

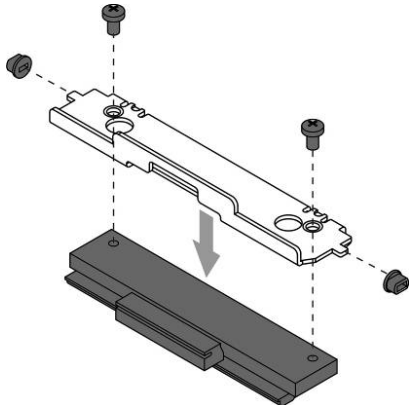
	Disassemble the two latch tension springs using the nipper pliers, disassemble the two M2*5 pan head screws fixing the latch using the screw driver, pull out the latch shaft and take out the latch.
---	--

4.2.8 Disassembly of cutter sliding blade and print platen roller

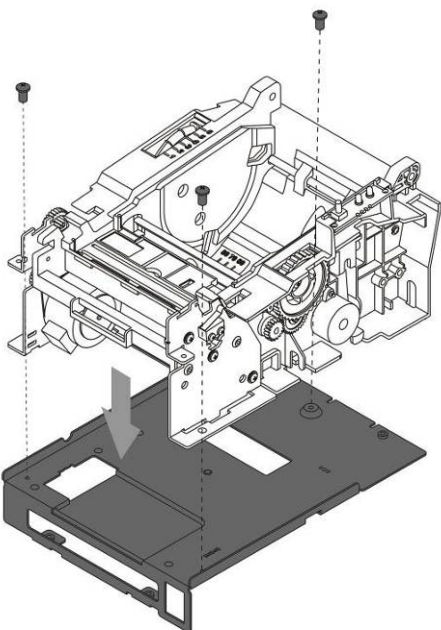
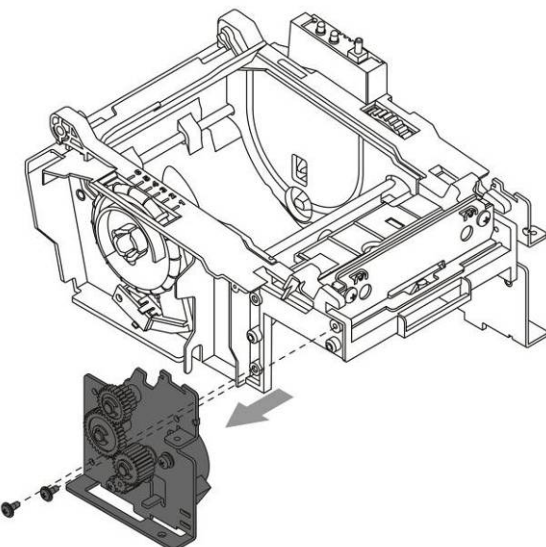
Figure	Description
	Take off the cutter sliding blade module with force in the direction shown in the figure without using any tools.
	Disassemble the four M2.6*6 special cutter screws using the screw driver and the left and right holders of cutter on both sides, and then take off the cutter sliding blade.
	Disassemble the two $\phi 3$ "E"-rings fixing the platen roller, take off the platen roller gear and shaft sleeves in order and finally take off the platen roller.

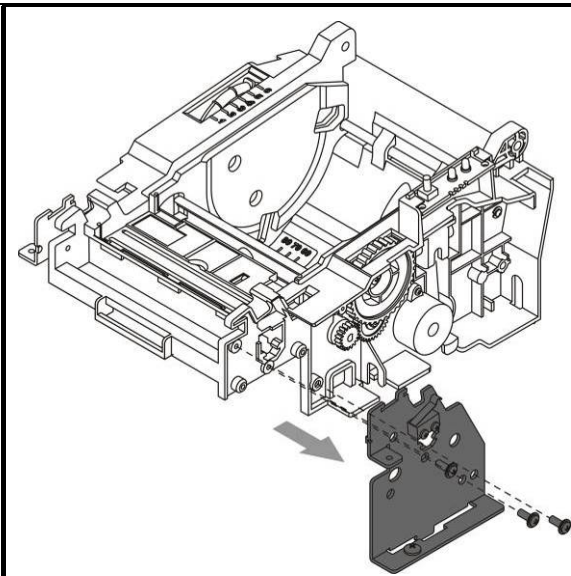
4.2.9 Disassembly of cutter fixed blade and print head

Figure	Description
	Push the cutter fixed blade module with force in the direction of the arrow in the figure, and take off the cutter fixed blade module from the printer.
	Disassemble the two M3*8 screws shown in the figure and take off the cutter fixed blade.
	Unplug the print head cable, and take off the print head module in the direction of the arrow.

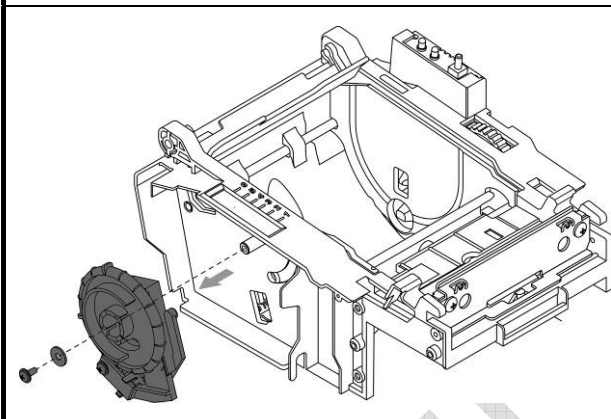
	Disassemble the two M3*5 screws fixing the print head and take off the print head.
---	---

4.2.10 Paper cabinet disassembly

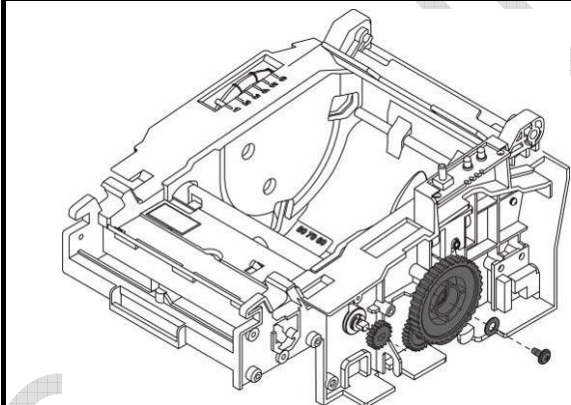
Figure	Description
	Disassemble the three M3*5 screws fixing the paper cabinet module and take off the cover plate of main board on the paper cabinet module.
	Disassemble the two ST2.6*6 self-tapping screws fixing the transmission module and take off the transmission module.



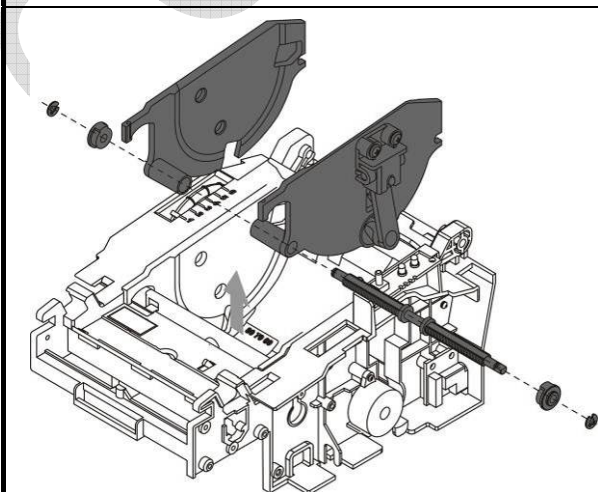
Disassemble the three ST2.6*6 self-tapping screws fixing the right support plate module, and take off the right support plate module.



Disassemble the ST2.6*6 self-tapping screw and $\phi 3$ "E"-ring and take off the paper near end sensor module.

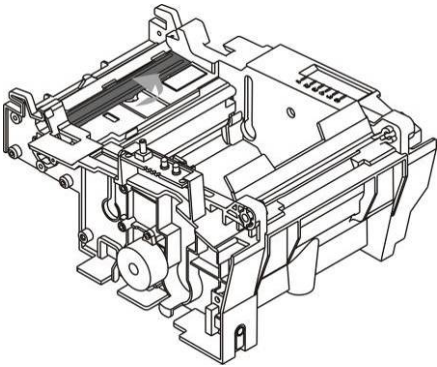
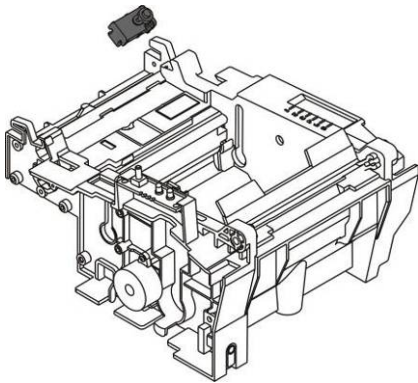


Disassemble the ST2.6*6 self-tapping screw in the figure and take out the paper cabinet baffle adjusting thumb wheel and transmission gear module.

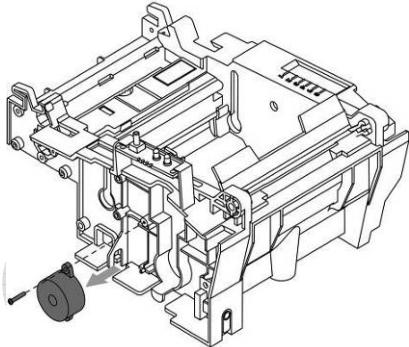


Rotate the adjusting screw arbor gears to adjust the left and right paper guides in the paper cabinet to about 60mm level. Disassemble the adjusting screw arbor gears and two $\phi 3$ "E"-rings on both sides, take off the shaft sleeves on both sides and paper guide module.

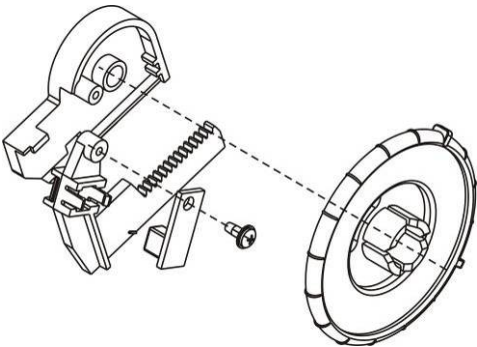
4.2.11 Disassembly of paper sensor / mark sensor

Figure	Description
	Disassemble the transparent cover of lower mark sensor.
	Slide the lower mark sensor to the right and take off the sensor from above.

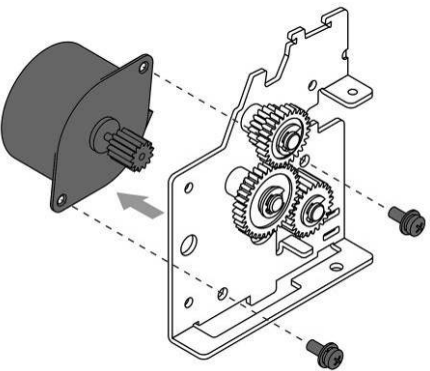
4.2.12 Buzzer disassembly

Figure	Description
	Disassemble the ST1.6*8 self-tapping screw fixing the buzzer and $\phi 2$ spacer, and take off the buzzer from the paper cabinet.

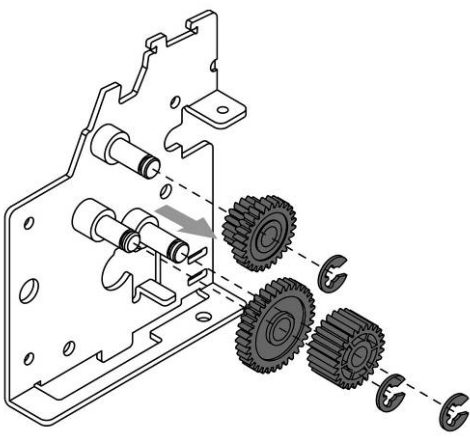
4.2.13 Paper near end sensor disassembly

Figure	Description
	Disassemble the ST2.6*6 self-tapping screw fixing the paper near end sensor, and take off the paper near end sensor from the sensor holder.

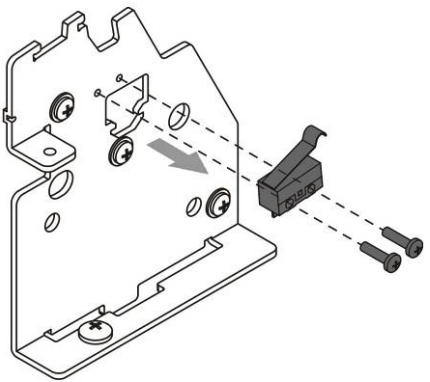
4.2.14 Motor disassembly

Figure	Description
	Disassemble the two M3×8 combination screws fixing the motor and take off the motor from the transmission support plate.

4.2.15 Gear disassembly

Figure	Description
	Disassemble the three $\phi 4$ "E"-rings fixing the gears and take off the gears from the transmission support plate.

4.2.16 Micro switch disassembly

Figure	Description
	Disassemble the two M2*8 pan head screws fixing the micro switch and take off the micro switch.

4.3 Assembly

Assemble the printer in the reverse sequence of disassembly.

5 Routine maintenance

Caution:

- ✧ Before starting routine maintenance, ensure that the printer power is turned off;
- ✧ Do not use solvents like gasoline or acetone;
- ✧ When cleaning sensors, the printer should not be powered on until the pure alcohol has completely evaporated.
- ✧ Do not use hand or metal object to touch the print head surface, and do not scratch the surface of print head, print platen roller or sensors using tools like forceps.
- ✧ It is recommended that the maintenance cycle should be no longer than one month.

5.1 Cleaning print head and platen roller

When any of the following cases occurs, the print head and platen roller should be cleaned:

- Printout is not clear;
- Some column on the page is not clear;
- Paper feeds or retracts with big noise.

Follow the steps below to clean the print head and platen roller:

- 1) Turn off the printer and open the top cover;
- 2) If the printing was just finished, please wait for the print head to cool down completely;
- 3) Wipe off the dust and stains on the surface of print head and platen roller with alcohol cotton (it should be wrung out);
- 4) After the alcohol is completely evaporated, close the top cover.

5.2 Cleaning paper end/mark sensor

When any of the following cases occurs, the paper end/mark sensor should be cleaned:

- During the printing, the printer stops printing once in a while and reports paper end;
- Marks cannot be identified effectively;
- No alarm for paper end.

The cleaning steps are as follows:

- 1) Turn off the printer;
- 2) Press the cover open lever to open the top cover of the printer;
- 3) Take off the transparent sensor cover;
- 4) Use soft cotton cloth dipped with pure alcohol (wrung out) to wipe off the dust and stains on the surface of the sensor;
- 5) Wait until the pure alcohol has completely evaporated, then close the top cover to end sensor cleaning.

6 Troubleshooting

Refer to this chapter for solution when a problem occurs. If the problem still cannot be solved, please contact your local dealer or manufacturer for assistance.

6.1 Troubleshooting

Malfunction	Causes	Solution
Error LED flashes 2 times	No thermal paper or thermal paper used up	Replace or install thermal paper.
	Flash when paper is present, paper sensor broken	Replace paper sensor.
Error LED flashes 3 times	Top cover is lifted up.	Close the top cover.
	If top cover is not lifted up, then the micro switch is broken.	Replace the micro switch.
Error LED flashes 4 times	Cutter error	Press the button momentarily to clear the cutter error.
	Cutter error cannot be cleared through pressing the button.	Replace the cutter.
	If the error still cannot be cleared after the cutter is replaced and PIN 1 and 2 of CNCUT connector are short-circuited to earth, then the drive chip ASMTC6301B(U401) is broken.	Replace the chip.
Error LED flashes 5 times	Power voltage abnormal	Check if the output voltage of power adapter is normal or not.
Error LED flashes 6 times	Print head cable is loose or not connected.	Check the connector.
	Print head overheating alarm	Stop printing, and wait for the print head to cool down.

Table 6.1-1 printer error types

6.2 Printer actions when an error occurs

- Stop printing;
- Error LED flashes;
- Buzzer beeps;

6.3 Printer status description

- **Error status**
Including print head lift-up, paper end, print head overheating, mark calibration error, cutter error, printer voltage abnormal, etc.;
- **Standby status**
Printer has no error, is not printing or feeding paper, and is waiting for a print task;
- **Print status**
The process from starting printing to finishing printing the print content;
- **Cutting/tearing status**
The process of cutter cutting paper/user tearing paper;

Caution:

- The printer status information can be inquired through command or ASB.
- Printer status can be inquired through the following two methods:

- 1) Use status inquiry command (real-time command):
 - After the user sends the status inquiry command to the printer, the printer will return its status through interface in fixed data format. The user can inquire the current status of the printer through the data returned from the printer.
 - For the details of status inquiry command, refer to Programming Manual.
- 2) Inquiring printer status through ASB function:
 - If the ASB function is enabled, the printer will return the current status of the printer through interface automatically.

CONFIDENTIAL

Appendix

Appendix 1 command list

Refer to RP-600 Programming Manual.doc.

Appendix 2 EEPROM configuration table

Printer parameters can be configured through EEPROM, and the related addresses and definitions of EEPROM configuration are as shown in the table below:

Address	Explanation	Remark
14H	Set CR command mode	0:CR command disabled 1:CR command enabled
16H	Set baud rate of serial interface	0:9600bps 1:19200bps 2:38400bps 3:57600bps 4:4800bps 5:2400bps 6:1200bps 7:115200bps
17H	Set number of serial data bits	7:7 data bits 8:8 data bits
19H	Set stop bits of serial interface	1:1 stop bit 2:2 stop bits
1BH	Set parity of serial interface	0:none 1:odd 2:even
4AH	Default print right margin	Value range: 0~9, unit: mm
4EH	Default print left margin	Value range: 0~9, unit: mm

Table appendix 2-1

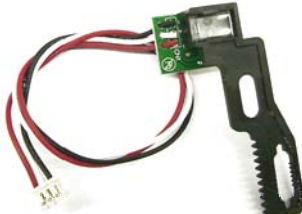
EEPROM configuration of printer can be realized through command as follows:

【Format】 ASCII ESC s B E a1 a2 data1 data2 n
 Hex 1B 73 42 45 92 9A data1 data2 n
 Decimal 27 115 66 69 146 154 data1 data2 n

【Range】 0≤n≤251
 a1 = 92h
 a2 = 9Ah

【Description】 Write WORD type of data into EEPROM, address specified by n. data1 is low byte data, and data2 is high byte data;
 After changing the EEPROM configuration, the new configuration cannot come into effect until the printer is restarted.

Appendix 3 list of easily damaged parts

Part	No.	Qty	Reference picture
Main board	—	1	
Paper end/mark sensor module	7600-9031301	1	
Paper near end sensor module	7600-9010972	1	
Feed button and LED module	7600-9031300	1	
Micro switch module	7600-9001092	1	

Power switch module	7600-9010792	1	
Power switch module	7600-9010793	1	
Paper feed motor	3200-9005397	1	
Cutter	3100-9028140	1	
Print head cable	7110-9000369	1	
Platen roller gear (18)	8203-9007107	1	

Transition gear (30)	8203-9000594	1	
Gear (16-35)	8203-9001880	1	
Gear (25)	8203-9000993	1	
Gear (18-27)	8203-9001777	1	
Platen roller gear (26)	8203-9002239	1	
Paper guide adjusting thumb wheel	8206-9016575	1	

Platen roller	8.30.1.9018904	1	
---------------	----------------	---	--

CONFIDENTIAL

Appendix 4 printer exploded view

1. Exploded view of printer

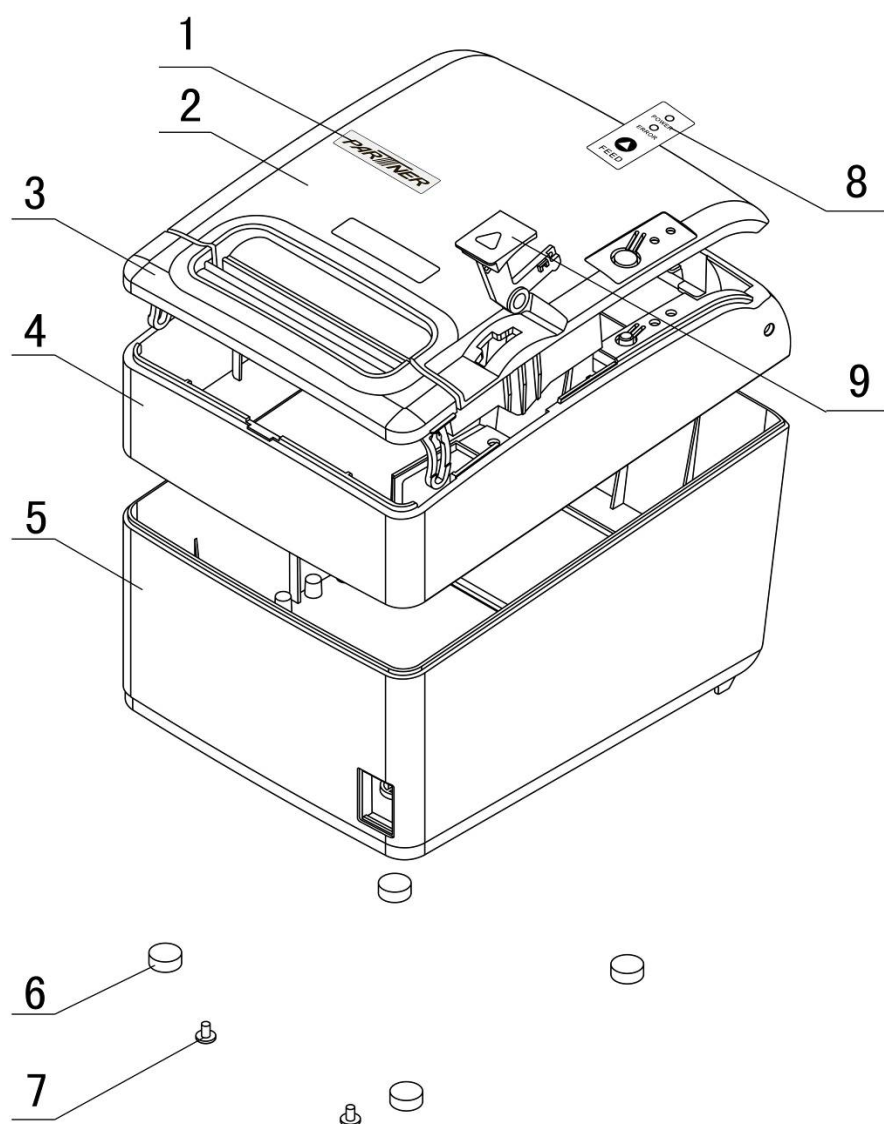


Figure appendix 4-1

2. Exploded view of mechanism

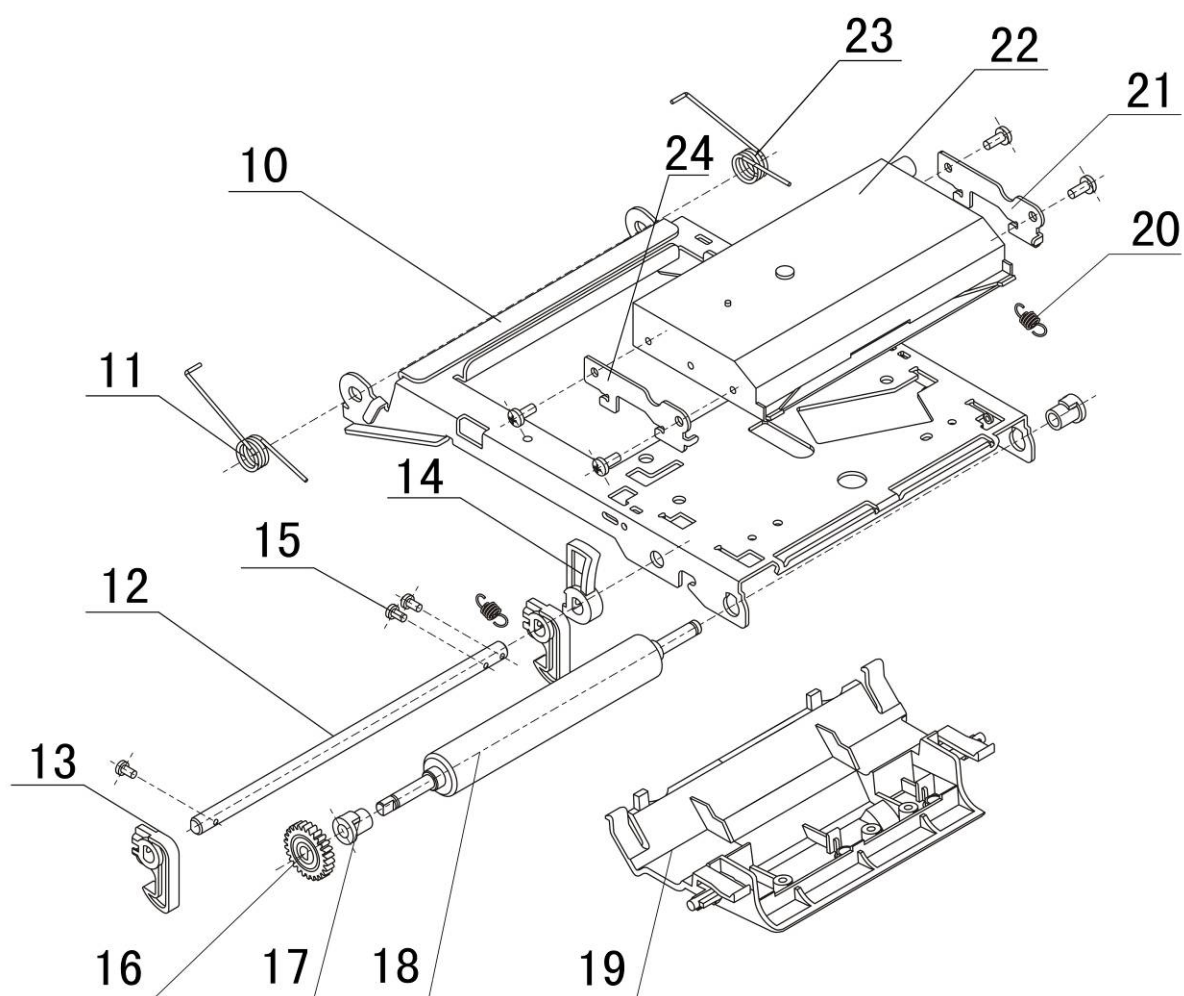


Figure appendix 4-2

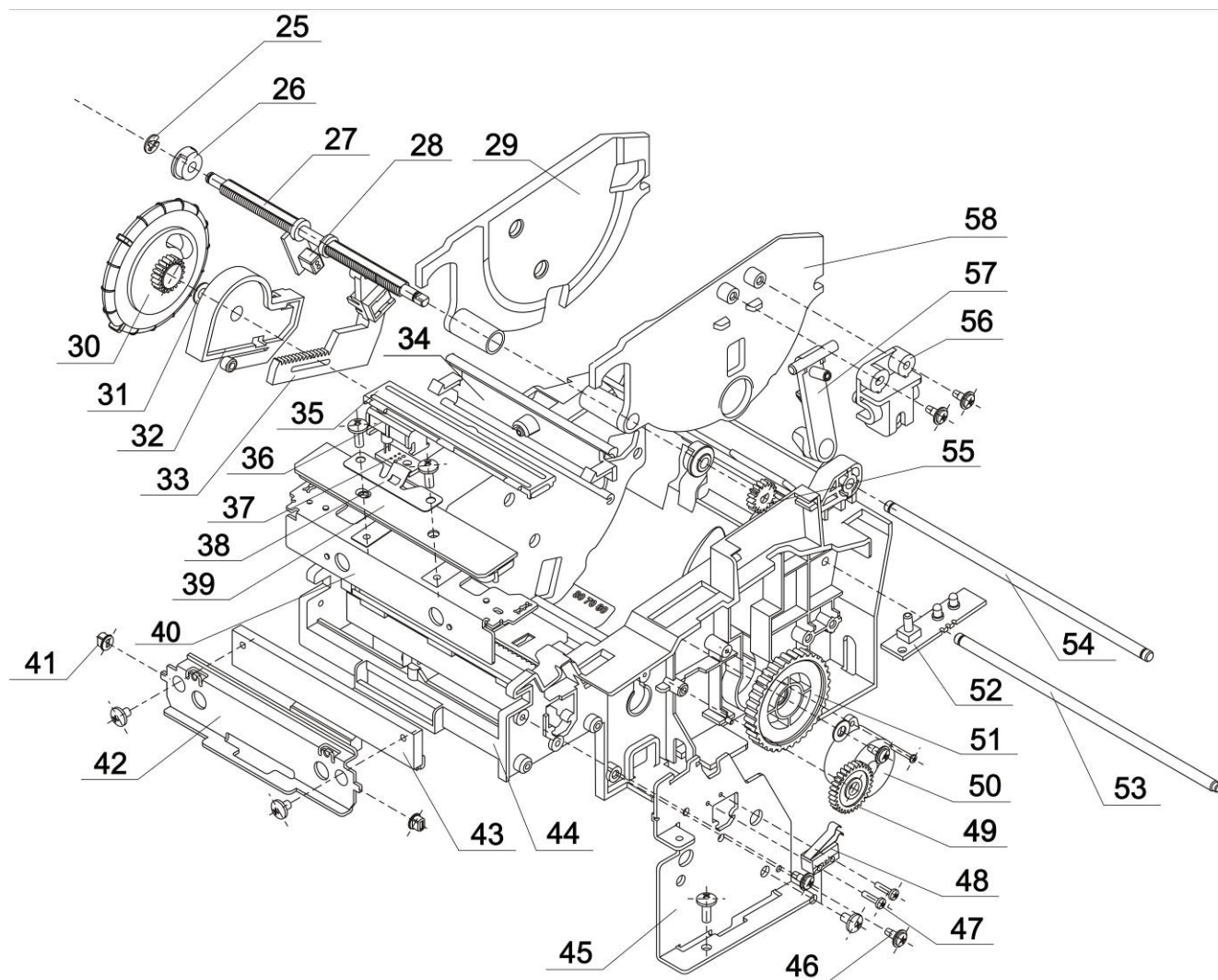


Figure appendix 4-3

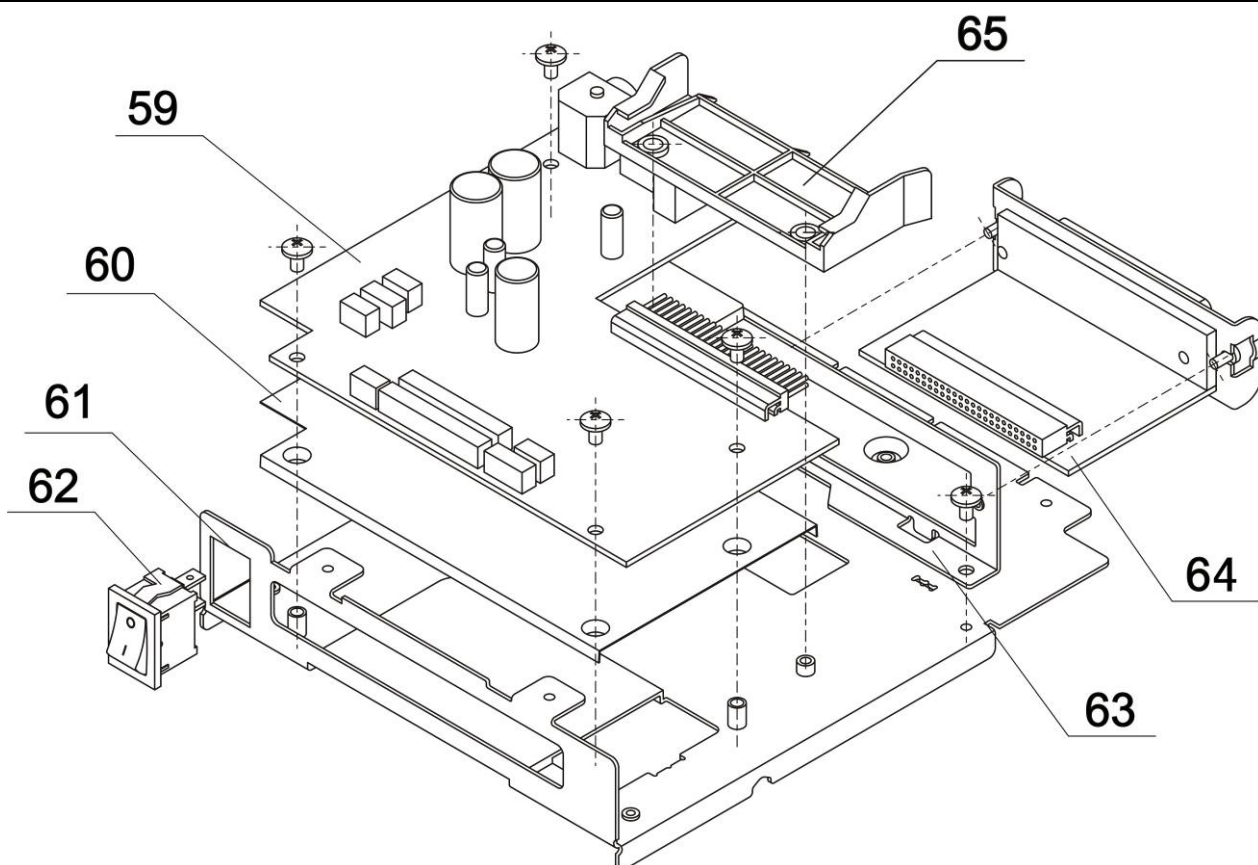


Figure appendix 4-4

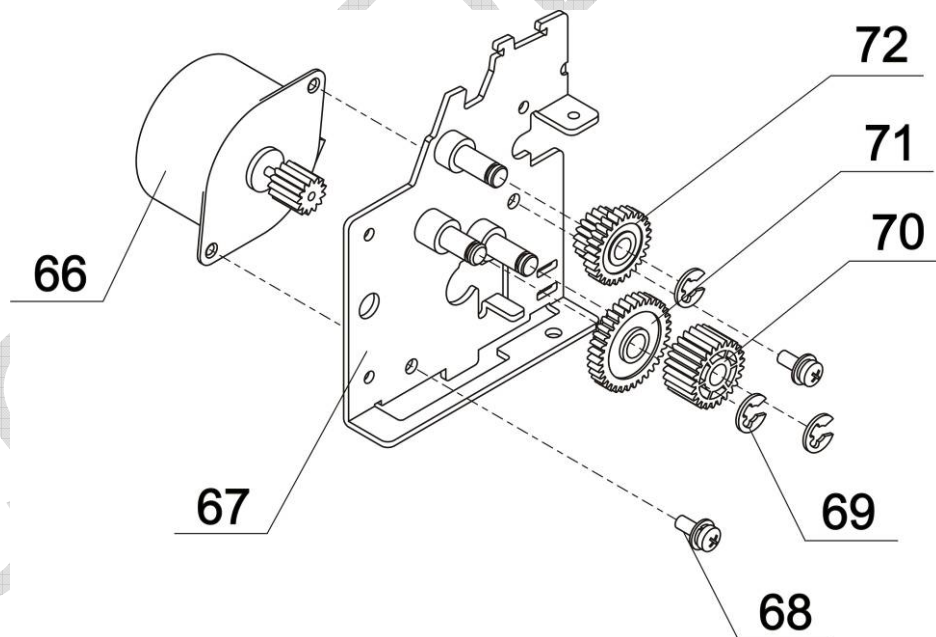


Figure appendix 4-5

Appendix 5 parts list

No.	Part No.	Part name	Qty	Remark
1	8.01.2.9031280	Trademark label (PARTNER)	1	
2	8.20.1.9031268	Top cover	1	White
	8.20.1.9031267			Black
3	8.20.1.9031274	Cutter cover	1	White
	8.20.1.9031273			Black
4	8.20.1.9031272	Middle cover	1	White
	8.20.1.9031271			Black
5	8.20.1.9031276	Bottom cover	1	White
	8.20.1.9031275			Black
6	8.30.3.9001997	Rubber foot	4	
7	4.00.6.9000295	Spherical cross screw, M3×5	13	
8	8.01.2.9031279	Button label	1	
9	8.20.1.9031270	Cover open lever	1	White
	8.20.2.9031269			Black
10	8.10.3.9020774	Top cover holder	1	
11	8.00.5.9020789	Top cover torque spring 1	1	
12	8.00.1.9018882	Top cover shaft	1	
13	8.21.9.9018879	Latch	2	
14	8.20.2.9018880	Mechanism handle	1	
15	4.00.0.9011260	Pan head screw, M2×5	3	
16	8.20.3.9002239	Platen roller gear (26)	1	
17	8.21.1.9001040	Shaft sleeve	2	
18	8.30.1.9018904	Platen roller	1	
19	8.20.7.9020778	Upper mark sensor holder	1	
20	8.00.5.9003403	Handle tension spring	2	
21	8.10.3.9018895	Right holder of cutter	1	
22	3.1.00.9028140	Cutter	1	
23	8.00.5.9020790	Top cover torque spring 2	1	
24	8.10.3.9018896	Left holder of cutter	1	
25	4.40.0.9000293	"E"-ring, Φ3.0	6	
26	8.21.1.9001307	Shaft sleeve 1	2	
27	8.20.7.9018881	Adjusting screw arbor	1	
28	7.6.00.9010972	Paper near end sensor		
29	8.20.7.9020765	Left paper guide	1	
30	8.20.2.9020768	Paper near end adjusting thumb wheel	1	
31	4.20.5.9005281	Large gasket, Φ3, outside diameter Φ9	1	
32	8.20.7.9010001	Paper near end module holder	1	
33	8.20.7.9018425	Paper near end sensor holder	1	

34	8.20.7.9018871	Buffer plate	1	
35	8.21.5.9018873	Lower mark sensor cover	1	
36	8.20.7.9018874	Lower mark sensor seat	1	
37	7.6.00.9031301	Mark sensor	1	
38	8.10.4.9018900	Tear-off bar ground flat spring	1	
39	8.20.1.9005382	Fixed blade backing plate	1	
40	8.10.3.9020773	Fixed blade fixing plate	1	
41	8.21.1.9014484	Print head fixing plate sheath	2	
42	8.10.3.9031266	Print head fixing plate	1	
43	3.0.00.9022537	Print head	1	
44	8.20.7.9020764	Paper cabinet	1	
45	8.10.3.9020771	Right support plate	1	
46	4.00.2.9000301	Pan head self-tapping screw, ST2.6×6-F-H	11	
47	4.00.0.9000057	Pan head screw, M2×8	2	
48	7.6.00.9001092	Micro switch	1	
49	8.20.3.9000594	Transition gear (30)	1	
50	0.4.10.9015006	Buzzer	1	
51	8.20.6.9016575	Paper guide adjusting thumb wheel	1	
52	7.6.00.9031300	Button PCB	1	
53	8.00.1.9018429	Paper cabinet shaft	1	
54	8.00.1.9018882	Top cover shaft	1	
55	8.20.3.9007107	Platen roller gear (18)	1	
56	8.20.7.9018868	Paper roll eject-plate holder	1	
57	8.20.7.9018867	Paper roll eject-plate	1	
58	8.20.7.9020766	Right paper guide	1	
59	—	Main control board	1	
60	8.00.6.9020795	Main control board protective film	1	
61	8.10.2.9020777	Main board cover plate	1	
62	7.6.00.9010792 7.6.00.9010793	Power switch	1	White Black
63	8.10.3.9018892	Interface guard board	1	
64	—	Interface board	1	
65	8.20.1.9003328	Guide plate	1	
66	3.2.00.9005397	Motor	1	
67	7.70.0.9031262	Transition support plate riveting module	1	
68	4.01.3.9000297	Combination screw, M3×8	2	
69	4.40.0.9000627	“E”-ring, Φ4.0	3	
70	8.20.3.9000993	Gear (25),	1	
71	8.20.3.9001880	Gear (16-35)	1	
72	8.20.3.9001777	Gear (18-27)	1	

Appendix 6 overall size (unit: mm)

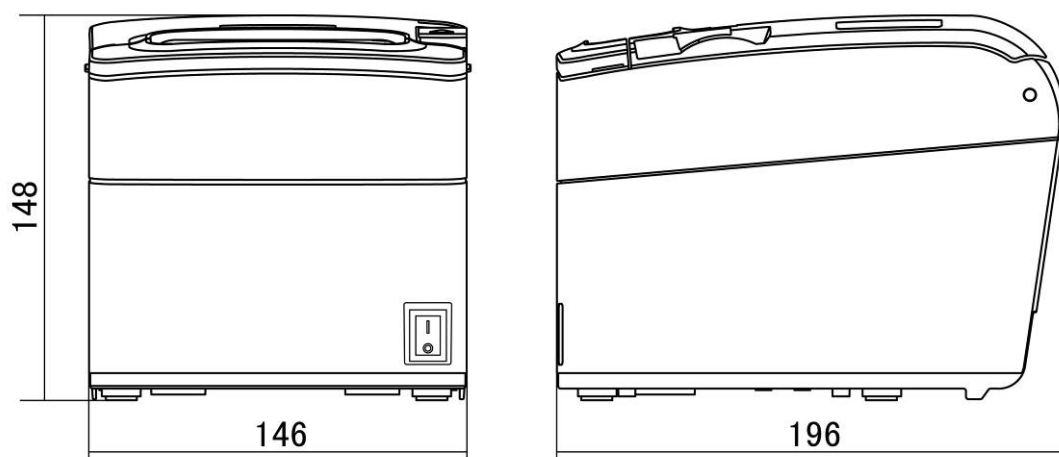


Figure appendix 6-1

Appendix 7 silk-screen drawing of main control board

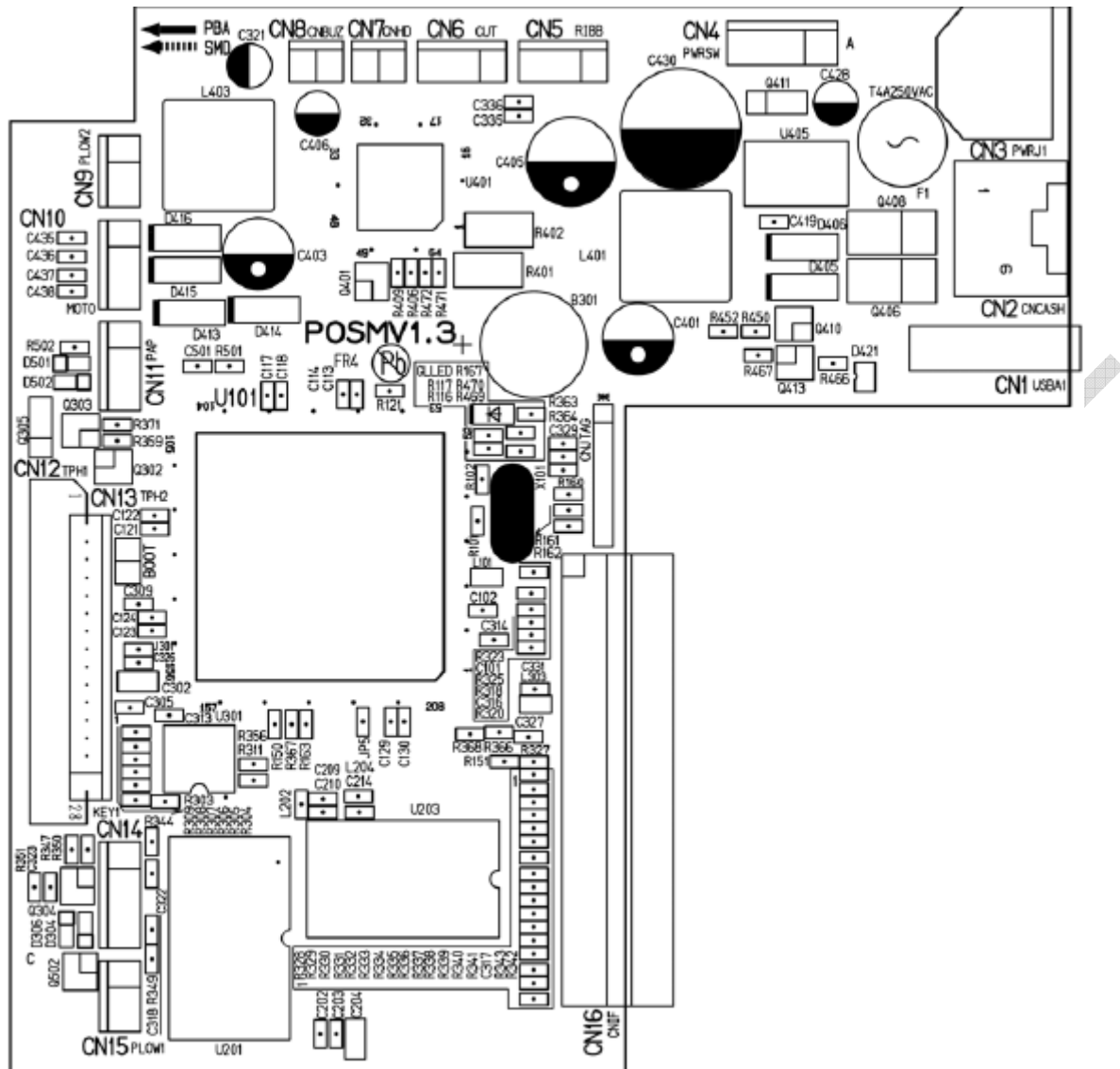


Figure appendix 7-1 front layout of main control board

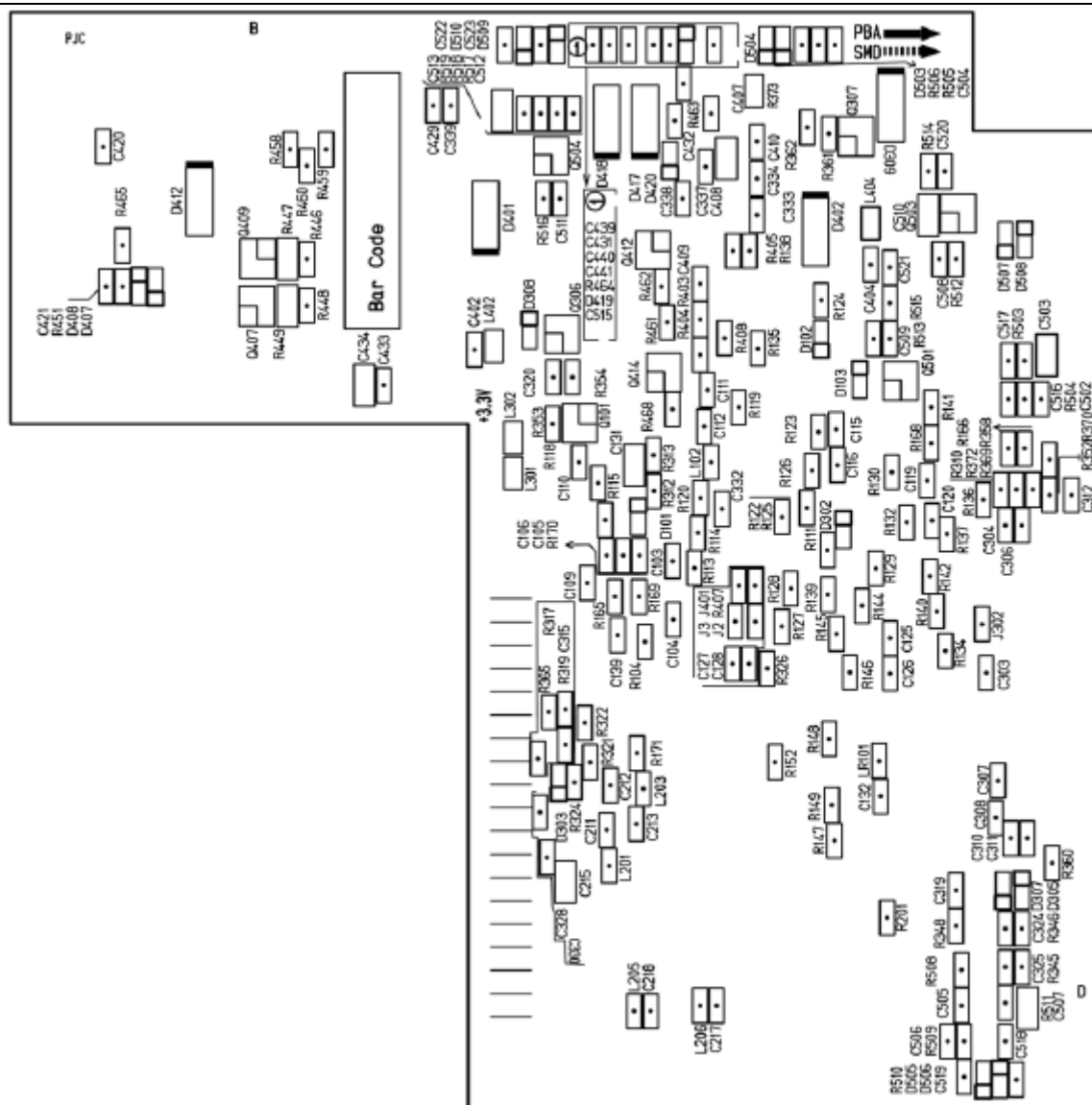


Figure appendix 7-2 back layout of main control board

Appendix 8 button configuration list

PARAMETER SETTING BY FEED BUTTON					
MAIN MENU					
Exit	->1				
Print Self Test	->2				
Configuration	->3				
		CONFIGURATION			
		Exit Without Save	->1		
		Exit With Save	->2		
		Communication	->3	Back To Last Menu	->1
				USB Interface	->2
				USB MODE:API MODE	
				Back To Last Menu	->1
				WinDriver Mode	->2
				API Mode	->3
				Serial Interface	->3
				SERIAL INTERFACE	
				Back To Last Menu	->1
				Baud Rates	->2
				BAUD RATES:115200bps	

					Back To Last Menu	->1
					9600bps	->2
					19200bps	->3
					38400bps	->4
					57600bps	->5
					4800bps	->6
					2400bps	->7
					1200bps	->8
					115200bps	->9
			Parity	->3	PARITY:NONE	
					Back To Last Menu	->1
					None	->2
					Odd	->3
					Even	->4
			Data Bits	->4	DATA BITS : 8 Bits	
					Back To Last Menu	->1
					7 Bits	->2
					8 Bits	->3
			Stop Bit(s)	->5	STOP BITS: 1 Bit	
					Back To Last Menu	->1
					1 Bit	->2
					2 Bits	->3
			Handshaking	->6	HANDSHAKING:DTR/DSR	
					Back To Last Menu	->1
					DTR/DSR	->2
					XON/XOFF	->3
			Data Receive Error	->7	DATA ERROR SETTING: Print '?'	
					Back To Last Menu	->1
					Ignored	->2
					Print '?'	->3
		Rx Buff Size	->4	RX BUFFER SIZE:64K Bytes		
				Back To Last Menu	->1	
				4k Bytes	->2	
				45 Bytes	->3	
				64K Bytes	->4	
Mechanism & Hardware	->4	HARDWARE SETTINGS				
		Back To Last Menu	->1			
		Mark Sensor	->2	MARK SENSOR: Disable		
				Back To Last Menu	->1	
				Enable	->2	
				Disable	->3	
		Cutter	->3	CUTTER:		
				Back To Last Menu	->1	
				Cut Mode	->2	CUT MODE SETTINGS:

			Settings			Default CutMode		
					Back To Last Menu	->1		
					Enable	->2		
					Disable	->3		
					Full Cut Mode	->4		
					Partial Cut Mode	->5		
					Default Cut Mode	->6		
			Auto Cut Settings		->3	AUTO CUT SETTINGS: No use this function		
					Back To Last Menu	->1		
					Cut paper when cover is closed	->2		
					No cut paper when cover is closed	->3		
					Cut paper when power on	->4		
					No cut paper when power on	->5		
					Disable	->6		
		Buzzer	->4	BUZZER: Normal Volume				
				Back To Last Menu	->1			
				Low Volume	->2			
				Normal Volume	->3			
				High Volume	->4			
				Higher Volume	->5			
				Highest Volume	->6			
				Disabled	->7			
		Power Supply	->5	POWER SUPPLY: Normal				
				Back To Last Menu	->1			
				Normal	->2			
				Low Power Mode	->3			
	Print Settings	->5	PRINT SETTINGS					
			Back To Last Menu	->1				
			Darkness Settings	->2	DARKNESS SETTING: Normal			
				Back To Last Menu	->1			
				Low	->2			
				Normal	->3			
				High	->4			
				Extra High	->5			
		Paper Roll Width	->3	PAPER ROLL WIDTH:80.0mm				
				Back To Last Menu	->1			
				54~59mm	->2	Select A Width		
					Back To Last Menu	->1		
					54mm	->2		
					55mm	->3		

				56mm	->4
				57mm	->5
				58mm	->6
				59mm	->7
			60~65mm	->3	Select A Width
				Back To Last Menu	->1
				60mm	->2
				61mm	->3
				62mm	->4
				63mm	->5
				64mm	->6
				65mm	->7
			66~71mm	->4	Select A Width
				Back To Last Menu	->1
				66mm	->2
				67mm	->3
				68mm	->4
				69mm	->5
				70mm	->6
				71mm	->7
			72~77mm	->5	Select A Width
				Back To Last Menu	->1
				72mm	->2
				73mm	->3
				74mm	->4
				75mm	->5
				76mm	->6
				77mm	
			78~82mm	->6	Select A Width
				Back To Last Menu	->1
				78mm	->2
				79mm	->3
				80mm	->4
				81mm	->5
				82mm	->6
			Left Margin	->4	LEFT MARGIN:4mm
				Back To Last Menu	->1
				0mm	->2
				1mm	->3
				3mm	->4
				5mm	->5
				7mm	->6
				9mm	->7
			Right Margin	->5	RIGHT MARGIN:4mm
				Back To Last Menu	->1
				0mm	->2
				1mm	->3
				3mm	->4
				5mm	->5
				7mm	->6

				9mm	->7	
		CR Command	->6	CR COMMAND: Disable		
				Back To Last Menu	->1	
				Enable	->2	
				Disable	->3	
		Code Page	->7	CODE PAGE SETTING		
				Back To Last Menu	->1	
				Print all codepages	->2	
				Select a codepage	->3	
		Save Paper Level	->8	SAVE PAPER LEVEL: Disabled		
				Back To Last Menu	->1	
				Disable	->2	
				25%	->3	
				50%	->4	
				75%	->5	
				100%	->6	
		Two-color Verify	->9			
		Gray Bmp Verify	->10			
Paper Sensor Settings	->6	PAPER NEAR END SETTINGS				
		Back To Last Menu	->1			
		Paper Low Alarm	->2		PAPER LOW ALARM: Enabled	
			Back To Last Menu		->1	
			Enable		->2	
			Disable		->3	
		Stop Print When PAPER Low	->3		STOP PRINT WHEN PAPER LOW: Disable	
			Back To Last Menu		->1	
			Enable		->2	
			Disable		->3	
		Paper Near End Sensor	->4		PAPER NEAR END SENSER: Enabled	
			Back To Last Menu		->1	
			Enable		->2	
			Disable		->3	
		Set Default Config	->7		SET DEFAULT CONFIGURATION	
					Back To Last Menu	->1
Set Printer To Default Configuration	->2					
FONTA/FONTB Settings	->8	Current Font:FONTA				
		Back To Last Menu	->1			

		Select FONTA	->2			
		Select FONTB	->3			
		Select UDFONTA	->4			
		Select UDFONTB	->5			
	Beep settings	->9	Beep settings:Disabled			
		Back To Last Menu	->1	BEEP MODE		
		Enable External Herald	->2			
		Enable Internal buzzer	->3			
			Back To Last Menu			->1
			Mode 1			->2
			Mode 2			->3
	Mode 3		->4			
	Mode 4		->5			
		Mode 5	->6			
		All Beep disabled	->4			
Enter code,then hold Button Down at least 1 second to validate						
Sensor Test	->4	Sensor Test Mode: ERROR LED state will change according to sensor state To EXIT, hold button down at least 1 second				
Cutter Test	->5					
Print Statistics	->6	RP-600 STATISTICS				
		TCUT	:0			
		TLFS	:0			
		ONTIME	:0			
Calibration	->7					
E05 Configuration	->8	E04CONFIGURATION				
	Exit Without Save	->1	Saving config, please wait. Configurations have been saved, please repower the printer			
	Exit With Save	->2				
	Reset JK-E05 Config	->3				
	Print Settings	->4				
		IP Address: MAC Address: SUBNET Mask: GATEWAY: Print Port: DHCP:				